

Procedure for locating oil and gas wells in the Appalachian Basin

Matthew D. Reeder^{1,2}; James I. Sams III^{1,2}; Raghava Gorantla^{1,2}; Natalie J. Pekney¹

¹National Energy Technology Laboratory, 626 Cochran Mill Road, Pittsburgh, PA 15236, USA

²NETL Support Contractor, 626 Cochran Mill Road, Pittsburgh, PA 15236, USA

Corresponding author: matthew.reeder@netl.doe.gov

Abstract

Locating undocumented (or poorly documented) oil and gas wells for environmental assessment is often difficult. Remnant features that confirm the presence of a well (intact casing/wellhead, well bore, *etc.*) are typically less than one meter in size and often obscured from direct observation on the ground or from the air (by dense vegetation, for example). In order to efficiently find such features, it is useful to first systematically compile publicly available digital data at progressively smaller scales prior to embarking on field campaigns. The information presented here describes the procedure developed and used by the U.S. Department of Energy's (DOE) National Energy Technology Laboratory (NETL) to locate potential oil and gas well sites for follow up field verification and characterization. Digital data are first compiled from national and state resources such as well location/production databases, historical topographic maps, historical aerial photographs, and Light Detection and Ranging (LiDAR) data. Although each data set is likely to be incomplete or inaccurate to some extent, combining the data resources using Geographic Information System (GIS) technology can generate potential well site targets with a higher degree of confidence which improves the efficiency of field work activities. This workflow was developed in the Appalachian Basin region and although certain aspects may be unique, the general process would be applicable to locating undocumented wells in other regions.

1 Introduction

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4 Current estimates from the US Environmental Protection Agency (EPA) National Greenhouse
5 Gas Emission and Sinks report puts the total number of abandoned (no longer actively
6 producing) oil and gas wells at nearly 3.7 million (EPA, 2023). Of these, recent studies suggest
7 that there are there are approximately 118,000 (Merrill *et al.*, 2023) to 123,000 (Boutot *et al.*,
8 2022) documented unplugged orphaned (no legally responsible owner of record) oil and gas
9 wells.

10 These wells can present a variety of environmental hazards, primarily by acting as a potential
11 conduit between the surface and subsurface allowing fluids (oil, brine, gas) to contaminate the
12 surface (Saint Vincent *et al.*, 2020a) and atmosphere (Kang *et al.*, 2014 and 2016; Pekney *et al.*,
13 2018). Additional environmental hazards such as subsidence or contamination of the subsurface
14 (Schout *et al.*, 2019) may not be immediately visible and can be further investigated once a well
15 is located.

16 The Bipartisan Infrastructure Law (BIL), passed in November 2021, includes funding through
17 the US Department of Interior to mitigate the environmental impacts of orphaned wells (White
18 House, 2022). As such, there is increased motivation by stakeholders (*e.g.* federal, state, tribal
19 agencies) in locating orphaned wells and quantifying methane leakage rates to facilitate plugging
20 and remediation efforts.

21 Locating orphaned wells within an area of interest may not be a straight-forward proposition,
22 however. Often, there are limits to the accuracy and or completeness of databases used by
23 stakeholders to keep track of oil and gas development activities and the combination of multiple

1 datasets has been shown to be beneficial in identifying orphaned wells (Saint Vincent *et al.*,
2 2020a, 2020b, and 2021).

3 The purpose of this manuscript is to introduce the methodology developed by the US Department
4 of Energy, National Energy Technology Laboratory (NETL) to locate orphaned wells within the
5 Appalachian Basin to help facilitate the collection of methane emissions measurements and other
6 future investigations by stakeholders. The Appalachian Basin has an extensive history of oil and
7 gas exploration dating back to the discovery of the Drake Well in 1859. The region presents a
8 variety of challenges in dealing with the legacy of the oil and gas industry, particularly in terms
9 of multiple episodes of development as well as challenging terrain and landscapes over a large
10 area.

11 The work presented here first goes through the progression of identifying and compiling publicly
12 available digital data and then describes the development of field data collection plans and
13 follow-up work and is meant to be a generic description of the process and not necessarily
14 specific to a particular site. As such, this methodology should be applicable to any area of
15 interest where digital data are publicly available.

16 17 18 **Methodology**

19 20 ***Digital Resources***

21 Once a project area has been selected, a review of historical activities at the site needs to be
22 considered. Questions such as “Does the area have a history of oil and/or gas development
23 activities?” and if so, “Are there known wells within the area of interest?” and “Has the ground
24 surface been significantly altered (*e.g.*, by logging or surface mining) that would obscure historical

oil and/or gas development in the area?” should be addressed. To answer these questions, publicly available digital data from state and national resources (complete list of resources in Appendix 1) can be compiled and processed using Geographic Information System (GIS) software (*e.g.*, ESRI ArcGIS or QGIS). The types of digital data that are typically available for this task are well location/production databases, topographic maps (current and historical), historical aerial photographs, and Light Detection and Ranging (LiDAR) data (Sams *et al.*, 2017; Pekney *et al.*, 2018; Saint Vincent *et al.*, 2020a, 2020b, and 2021).

Although each dataset listed above can provide location information for potential well sites, no individual dataset is guaranteed to be available, complete, or accurate (for example, a well may be identified on a topographic map but there are no corresponding database records). Compiling data from each available resource improves the overall confidence in identifying the location of any potential well sites in the project area.

Descriptions, benefits/drawbacks, and processing requirements for each resource are discussed below. It is important to note here that there is no specific order for assembling these data sources. The critical aspect at this stage of the process is to consider multiple resources in conjunction to generate an inventory of documented wells within the project area.

Well Location/Production Databases

Typically, each state has an entity that maintains records for oil and gas production activities including well-specific information on status, location, owner/operator, *etc.* Searchable national well databases have been compiled by the U.S. Geological Survey (USGS; Merrill *et al.*, 2023) and by private companies such as *Enverus* (formerly *DrillingInfo*). The data can usually be downloaded for the area of interest in spreadsheet or in a GIS compatible format (Figure 1). As

mentioned above, the information in these databases may not be complete or accurate, particularly for older wells. Well location data, for example, may be inaccurate or missing entirely. Errors in location can be in excess of 50 meters, which makes finding the site in the field very difficult (Saint-Vincent *et al.*, 2020b).

Topographical Maps

The U.S. Geological Survey (USGS) maintains a searchable online archive of topographical maps for the entire country. The locations of oil and gas wells are sometimes depicted on the maps along with major infrastructure and elevation data showing the terrain as contour lines (Figure 2). The maps are divided into quadrangles with the most detail on the 7.5-minute series (1:24,000 scale). These maps, referred to as digital raster graphics, can be downloaded in a GIS format.

Depending on the specific area of interest, historical topographic maps may be available from the 1940's through the 1980's. Additionally, there may be multiple versions of the same 7.5-minute map available based on how many times the map has been photo-revised (Moore, 2000). It is important to note that wells drilled after the latest revision up to present time will not be on the map, limiting the vintage of wells identifiable using this resource.

Using GIS software, a data layer on the location of wells shown on the topographic map can be created by manually digitizing the depicted locations (Figure 3). The accuracy of the digitized USGS point relative to an actual well identified in the field (if present) can be on the order of 20 m to 30 m (Saint-Vincent *et al.*, 2021). Note that surface features at well sites may have been altered after the latest map revision and there may no longer be any surface indication of the well's existence (Sams *et al.*, 2017).

Aerial Photography

The identification of potential well sites from aerial photography is more challenging but can yield beneficial information. Air photo coverage is available for most of the U.S. by national and state sources (Appendix 1). To obtain the most useful location data, the photos need to be manually georeferenced within the GIS. The accuracy of the georeferencing depends on the number and distribution of landmarks visible in both historic air photos and modern, georeferenced photos or maps. Factors such as digital resolution, film exposure, and flight parameters (altitude and angle) may limit the quality of the photographs available for a particular project area.

In forested areas, potential well sites can be spotted by looking for small open areas (no vegetation). A point may be digitized in the center of the open area and later field checked, typically with an accuracy of approximately 30 m (Figures 4 and 5). Other observable features such as haul roads assist in identifying potential well sites and aiding in navigation planning for field surveys.

Light Detection and Ranging (LiDAR)

LiDAR data (ground surface elevation points) can be very useful for locating potential well sites. Data processing within the GIS software generates high resolution ground surface images. Generally, the process involves converting the raw LiDAR data into a gridded (raster) dataset that shows the relative elevation of the ground surface. A cell size of one square meter (1 m²) is best for detecting unnatural surface features that resulted from oil and gas development activities. The GIS software can be consulted for guidance on using LiDAR data (such as [Use lidar in ArcGIS Pro—ArcGIS Pro | Documentation](#) for ESRI ArcGIS). Potential well sites are identifiable as

1 relatively flat rectangular areas with the long side perpendicular to the slope and parallel to the
2 contour of the terrain (Figure 6). An additional benefit of using LiDAR data is that haul roads and
3 trails are visible in much greater detail, allowing for more efficient ground reconnaissance route
4 planning.

5 LiDAR is a relatively new technology with a vast majority of the data collected within the last 20
6 years. As a result, if significant disturbance to the ground surface has occurred (e.g., logging,
7 surface mining, or agricultural activity) after the well was drilled, the well pad surface features
8 may no longer be identifiable.

9 10 *Other Historical Maps*

11 If available, documents such as company lease maps and/or farm-line maps may provide beneficial
12 information on the number and density of wells within a particular area of interest. Specific
13 locations for the wells would depend on how accurate the original data are and careful
14 georeferencing as described earlier.

15 In areas where there is a history of coal mining operations, underground mine maps may depict
16 wells that intersect the mine workings. Typically, the well locations would have been surveyed
17 with very high accuracy. These maps could be a particularly useful resource for areas where the
18 surface features of the wells may have been altered by subsequent land use. The US Department
19 of Interior – Office of Surface Mining Reclamation and Enforcement maintains a searchable
20 National Mine Map Repository (Appendix 1).

1 Compiling available digital data sources for the project area within a GIS application (Figure 7)
2 generates a resource that can be utilized to answer important questions, such as:

- 3 1) How many wells are located within the project area?
- 4 2) What is the spatial distribution of wells within the project area? Are there areas with
5 relatively higher/lower well density?
- 6 3) How do the location data agree among the sources for each well?
- 7 4) Are there wells that only show up in some of the sources?
- 8 5) Are there common well characteristics (such as age, depth/formation, oil vs gas
9 production, *etc.*) that can be gleaned from the combined data?
- 10 6) What are the physical characteristics of the project area? What type of terrain and
11 topography are typical? Are there any natural or man-made hazards/obstacles that
12 would prevent/permit access to the wells?
- 13 7) Are there any unnatural disturbances to the land surface?

1 ***Data analysis and field work planning***

2 The next phase of the well finding procedure involves analyzing the compiled digital data within
3 the framework of the project goals to identify what additional information is required and how to
4 collect the data most efficiently.

5 The main questions to consider are:

6 1.) Does every well within the project area need to be located or only a portion of the
7 wells?

8 2.) What information needs to be collected once a well is located?

9 3.) How big is the area of interest?

10 4.) What are the main topographic and terrain features of the project area?

11 5.) What is the estimated well density of the area of interest?

12 If only a portion of the potential wells within the project area need to be located (the stakeholder
13 has a limited project scope or lacks resources, for example), the compiled digital sources map
14 would be sufficient to allow for planning field work to locate the wells. However, if the goal of
15 the project is to locate all the wells within the area of interest, additional work (*i.e.*, identifying
16 undocumented wells) will be required to generate the necessary data.

17 The most efficient data collection method is highly dependent on the physical characteristics of
18 the project area. For small ($<0.25 \text{ km}^2$) areas of interest with relatively flat, open terrain, a
19 systematic walking survey (*e.g.* grid survey) could be conducted to identify any wells with intact
20 above-ground features not already accounted for in the compiled digital data. Wells that have had
21 the wellhead/casing removed would still be unidentifiable without technological aid, however.

For larger project areas, areas with inaccessible terrain, or areas where the ground surface has been altered significantly, magnetic surveys can be utilized to generate additional digital data and are described in greater detail below.

Magnetic Surveys

Magnetic surveys are accomplished by systematically passing a magnetometer (a sensor that measures magnetic field intensity) through the area of interest to identify abnormalities relative to the local background. These anomalies may be indicative of wells with steel casing. In order to determine what type of magnetic survey to employ, two questions must be addressed:

1) What is the size of the area where the survey will take place?

2) What are the topographical and terrain features of the area?

As before, the size of the area of interest is a significant factor in deciding what type of magnetic survey to pursue. A ground-based magnetic survey can be carried out in a small, relatively flat, open area by wearing a backpack-mounted magnetometer and external global positioning system (GPS). This method involves continuously walking parallel lines that are tightly spaced across the entire project area. In steep, uneven, densely vegetated areas it is often too strenuous and time-consuming to maintain the orientation or line-spacing required to generate useful data.

Airborne magnetic surveys can be used where walking surveys are impractical to conduct due to size or inhospitable surface conditions. Uncrewed Aerial Vehicle (UAV, a.k.a. “drone”) based surveys work well for areas of interest that are less than 5 km² in size. For drone surveys to be a viable option, there are site specific questions that need to be considered:

1) Are there vehicle accessible take-off/landing areas to stage the UAV system?

2) Will the UAV pilot be able to maintain Visual Line of Sight (VLOS) with the UAV from the staging areas?

3) Are there enough appropriate staging locations to cover the project area?

The map compiled from the digital data can be used to identify potential staging areas that would provide the necessary access for the UAV operations. Within the GIS software there are tools that can calculate and display the line of sight from a particular point on the map.

Project areas that are greater than 5 km² (which may be too large to survey in a cost-effective manner, potentially requiring multiple staging areas and additional personnel), or where a UAV survey is not a viable option because of the requirements described above, can be surveyed by means of a piloted helicopter system.

For both options, commercial vendors contracted to conduct the survey(s) will determine the flight parameters (flight height, speed, line spacing, and route) specific to the project area(s) to optimize the collection of useful data.

Once collected, the raw magnetic data are processed, typically by the vendor that conducted the survey, to identify locations of anomalies that may represent wells with steel casing within the project area. As with LiDAR data, the magnetic data are often presented as a raster image. The cells are color coded such that the magnetic anomalies can be identified easily, and the derived locations are added to the GIS map as an additional layer (Figure 8). The points from airborne surveys are normally within 30 m of the field verified well while points from ground-based surveys are most often less than 10 m.

Magnetic surveys cannot detect wells without steel casing; either wells with wood casing or other non-magnetic casing or wells where the steel casing was removed for recycle or reuse.

False positives can be generated by other magnetic materials buried in the area, but shallowly buried magnetic sources can be distinguished from well casing by the dipolar versus monopolar shape of the magnetic anomaly and the stronger upward continuance from vertical steel well

casing (Hammack *et al.*, 2020; Saint Vincent *et al.*, 2020b). False negatives can occur due to improper flight planning, particularly over areas with significant relief. An examination of airborne magnetic data suggests that complete well detection is only possible if adjacent surveys lines are no more than 50 m apart and the flight altitude must be 40 m above ground level or lower (Kaminski *et al.*, 2018). These issues demonstrate why combining magnetic surveys with the multiple existing data sets is optimal for successful well location. Further details about how to plan and execute aeromagnetic surveys, process the data, and conduct follow-up ground based efforts to verify the presence of a well site (and rule out false positives such as groundwater or mining wells, steel posts, bridges, culverts, and other discarded ferromagnetic objects) can be found in Sams *et al.*, 2017 and Hammack *et al.*, 2023.

It is worth noting here that recent work has demonstrated the application of a low-cost approach for magnetic surveys utilizing UAVs carrying a smartphone (Saneiyan and Mansourian, 2023) that may prove to be more accessible for stakeholders working in open areas.

Field Reconnaissance

The purpose of the steps above is to generate probable well locations as accurately as possible thus saving time and money by making the final field verification step much more efficient. The GIS map includes all the available well location data (databases, topographic, aerial photographs, LiDAR, and magnetic) for the project area. Using information from the topographical map and LiDAR, routes can be planned to efficiently navigate to the potential wells sites and look for

indications of development activities. Any field work should be conducted according to the organization's health and safety standards.

The compiled GIS map is exported to an interactive mapping and data collection application (Esri Field Maps, for example) on a smartphone or tablet. The application can use the internal GPS (accuracy of <5 m) of the device or can be connected wirelessly to an external high-resolution GPS (accuracy <1 m). Built-in cameras are useful for taking pictures that will automatically be matched with the recorded location. At minimum, a paper map and a hand-held GPS can be used for navigation and collecting location information for any located wells.

Field verification, a critical step in accurately identifying and characterizing oil and gas wells, involves navigating to the potential well sites to confirm the presence of a well by physically locating one of the following features: an intact wellhead, an open well casing in its original vertical orientation, or an open wellbore. Once a well is confirmed, data about the site is collected including a picture of the well and accurate GPS coordinates. In areas with limited sky exposure (such as densely forested areas or where there is steep terrain) the internal GPS may not have a strong enough signal to provide accurate positions in which case a follow-up trip with a more substantial external GPS receiver may be required. Marking located wells with flagging tape is recommended.

While field-checking the potential well sites, preliminary well characterization information can be gathered such as:

1) Is the well plugged or unplugged?

2) Is the well leaking oil, brine, or natural gas?

Although the plugging status of a well may have been included in a database, it is worthwhile to record the current condition of the well. Not all plugged wells were done following currently accepted practices and may still present an environmental hazard.

Wells that are leaking oil are generally identified as having black-stained soil and a distinct hydrocarbon odor. Often there is an area of dead vegetation surrounding the well.

Since natural gas (which is predominantly methane) leaking from a well is invisible and odorless, additional equipment is required to identify leaks (Table 1) and handheld methane sensors can be used to qualitatively detect the presence of methane. If an emission rate is required, the instruments needed to measure methane emissions are portable via backpack and measurements usually take less than 10 minutes per well (see Pekney *et al.*, 2018 and Saint-Vincent *et al.*, 2020a for a more detailed description of the procedure).

Additionally, it is worth noting the presence of oil and gas well related infrastructure (*e.g.* pumpjacks, pipelines, pull rods, anchors, storage tanks, *etc.*) at the site, particularly when a well cannot be confirmed as described above. The occurrence of one or more of these features is indicative of previous oil and gas development at the site.

Data collected from the field reconnaissance is added to the GIS either manually (from paper map/notebook) or automatically (from the mapping software) for analysis. After reviewing the data, additional field work could be required depending on the scope of the project. A workflow diagram summarizing the overall methodology described above is presented in Figure 9.

Table 1 List of field equipment options

Instrument	Description	Example

Smartphone/tablet	Records field data/pictures	iPhone; Android
High resolution GPS	Highly accurate (<1m) locations	Trimble R2; Trimble TDC600
Backpack magnetometer	Measures local magnetic field	GEM GSM-19
Qualitative methane detector	Identifies methane leak* (*detection limit provided by manufacturer)	Pergam Laser Methane Mini (LMm); Thermo Scientific TVA2020 FID/PID; Bascom-Turner Instruments Gas-Rover I and II; Heath RMLD-CS
Infrared Camera	Visual detection of methane leaks in real time	FLIR GF320
High flow sampler	Measures methane leakage rates	SEMTECH HI-FLOW 2

1

2

3 **Concluding Remarks**

4 Although every resource listed above may not be available for a particular area of interest, the
5 methodology described here demonstrates the utility of compiling multiple lines of evidence for
6 the location of oil and gas wells from various resources when the individual datasets may be
7 lacking in accuracy and/or completeness. This allows for more efficient planning (both in terms of

1 personnel-hours and cost) of field work activities for determining the physical locations, methane
2 emissions, wellbore integrity, and other environmental impacts of legacy oil and gas wells so they
3 can be prioritized for plugging and abandoning activities by State, Federal, and Tribal
4 stakeholders.

5 **Disclaimer**

6 This project was funded by the United States Department of Energy, National Energy Technology
7 Laboratory, in part, through a site support contract. Neither the United States Government nor any
8 agency thereof, nor any of their employees, nor the support contractor, nor any of their employees,
9 makes any warranty, express or implied, or assumes any legal liability or responsibility for the
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11 or represents that its use would not infringe privately owned rights. Reference herein to any
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13 otherwise does not necessarily constitute or imply its endorsement, recommendation, or favoring
14 by the United States Government or any agency thereof. The views and opinions of authors
15 expressed herein do not necessarily state or reflect those of the United States Government or any
16 agency thereof.

Appendix 1 – Digital Data Resources

National Resources

Well Databases

[Documented Unplugged Orphaned Oil and Gas Well Dataset | U.S. Geological Survey \(usgs.gov\)](#)

This database contains location information for known unplugged orphan wells compiled from state resources. The data are free to download in spreadsheet and GIS format.

[Well Database | Enverus \[frm. DrillingInfo\] \(Enverus.com\)](#)

[Well Database | S&P Global \[frm.IHS\] \(spglobal.com\)](#)

[Well Database | TGS \(tgs.com\)](#)

These private databases provide subscription-based services to access current and historical information on well locations, production, permit/construction dates, *etc.* The data can be exported for use in GIS software.

Historical Topographic Maps

[Historical Topographic Maps | U.S. Geological Survey \(usgs.gov\)](#)

This free service provides a range of historical and topographic maps at a variety of scales across the U.S. The exact date range available depends on the user's area of interest. The maps are accessed by searching for a location (area/town/quadrangle name, *etc.*) and selecting the age and scale of interest. A variety of file types are available for download for use in GIS applications.

Historical Aerial Photographs

[Historical Aerial Photography | U.S. Department of Agriculture \(usda.gov\)](#)

This fee-based service provides historical aerial photographs from 1955 to present for much of the U.S. Using the [ArcGIS Historical Availability of Imagery Interactive Map](#), once the county of interest is selected, a pop-up with a list of available products is shown and contains a link on how to order from the USDA website. Users can order via email and provide site location information (a shapefile of the project boundary, for example) to get the exact area required as a physical copy or digital download.

[Domestic Aerial Photography | National Archives \(archives.gov\)](#)

Aerial photographs prior to 1955 may be available from the National Archives. The above link gives a description on how to search for the county of interest to get a downloadable index with specific picture frames. Instructions on how to obtain digital copies from the National Archives is provided here: [Visit the Cartographic Research Room in College Park, Maryland | National Archives](#)

Historical Mine Maps

[Mine Map Repository | Office of Surface Mining Reclamation and Enforcement \(osmre.gov\)](#)

Comprehensive source of mine maps for the United States. Users can locate maps based on an area of interest and then send a request for the digital copy of the map.

State Resources for the Appalachian Basin

New York

[Well Database | NY Department of Environmental Conservation \(dec.ny.gov\)](#)

Free database with downloadable data on well status, production, location, depth, and owners/operators.

[GIS Clearinghouse | NY Office of Information Technology Services \(gis.ny.gov/gisdata\)](#)

This free webservice provides a wide variety of spatial data for download including [LiDAR](#) and [Orthoimagery](#) as well as state infrastructure, hydrography, and political boundaries, *etc.*

Pennsylvania

[PA Spatial Data Access \(PASDA\) | Pennsylvania State University \(pasda.psu.edu\)](#)

The state's public geospatial data clearinghouse with links to [well databases](#), [imagery](#), [historical photographs](#), [LiDAR](#), and [GIS data](#).

Ohio

[Well Database | OH Department of Natural Resources \(ohdnr.gov\)](#)

This database includes extensive well-specific information including location, status, permit/completion dates, and production, *etc.*

[GEOhio Spatial Information Portal | OH Department of Administrative Services \(ohio.gov\)](#)

The data are free to download and includes access to various geodata such as: imagery, digital elevation models, LiDAR, political boundaries, transportation, hydrography, *etc.*

West Virginia

[Well Database | WV Department of Environmental Protection \(dep.wv.gov\)](#)

This website provides a map and database that can be searched using the county of interest.

The data are free to download.

[LiDAR | WV GIS Technical Center \(wvu.edu\)](#)

This free interactive mapping tool provides access to download a variety of elevation data including LiDAR, Digital Elevation Models (DEMs), and high resolution contours.

[GIS Data | WV GIS Technical Center \(mapwv.gov\)](#)

This web portal provides a variety of GIS data for the state including imagery, elevation, boundaries, and infrastructure.

Kentucky

[Well Database | KY Geological Survey \(kgs.uky.edu\)](#)

This database (searchable by county) provides information about the well type, status, location, operator name (if available), drilling permits, etc. A [shapefile](#) containing the most up-to-date version of the KYGS database for the entire state is also available.

[Geospatial Data Clearinghouse | KY Division of Geographic Information \(ky.gov\)](#)

1 This website offers access to numerous downloadable GIS data sources including boundaries,
2 infrastructure, and geophysical information as well as [LiDAR](#) data.

3 4 *Tennessee*

5 [Well Database | TN Department of Environment & Conservation \(tn.gov\)](#)

6 This database has information about well type, status, location, depth, and operator, *etc.*

7
8 [GIS Data | TN Department of Finance and Administration \(tn.gov\)](#)

9 This website provides GIS resources for the state including infrastructure, [imagery](#), and
10 [LiDAR](#). The data are stored in a publicly accessible Google Drive and may require a Google
11 account to download.

12 13 *Alabama*

14 [Well Database | AL Oil and Gas Board \(ogb.alabama.gov\)](#)

15 This database has information about well type, status, location, and operator, *etc.*

16
17 [GIS Data | Geological Survey of Alabama \(gsa.state.al.us\)](#)

18 This web portal provides a variety of GIS data for the state including imagery, elevation,
19 boundaries, and infrastructure.

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